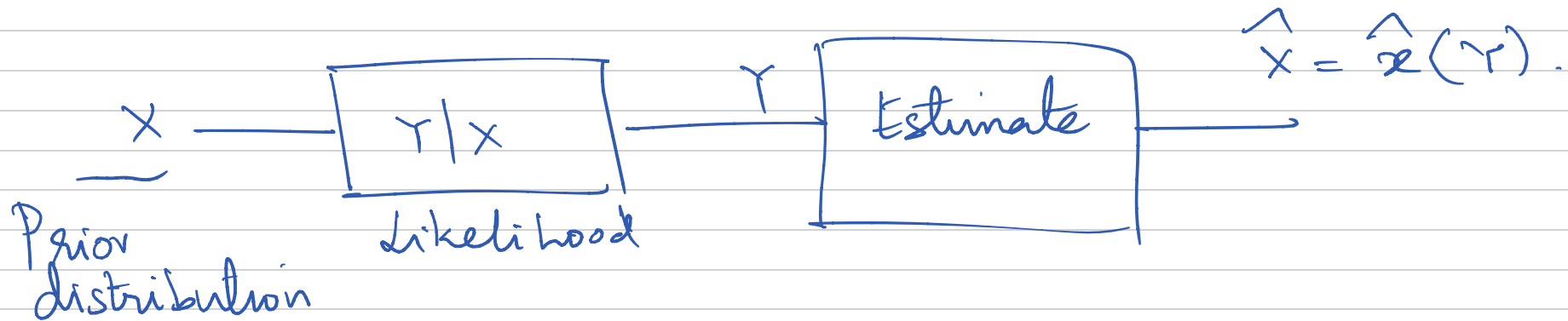


lec 38

Estimation



$P_{X|Y}(x|y)$: Posterior distribution

$f_{X|Y}(x|y)$

① X is discrete. We are interested in finding an estimator that minimizes probability of error

$$P(X \neq \hat{X}).$$

④ MAP estimator has minimum possible error probability

$$P(x \neq \hat{x}) \geq P(x \neq \hat{x}_{\text{MAP}}).$$

$$\hat{x}_{\text{MAP}} = \hat{x}_{\text{MAP}}(Y).$$

$$\hat{x}_{\text{MAP}}(y) = \arg \max_{x \in \mathcal{X}} P_{X|Y}(x|y).$$

$$= \arg \max_{x \in \mathcal{X}} \frac{f_{Y|X}(y|x) P_X(x)}{f_Y(y)}.$$

$$= \arg \max_{x \in \mathcal{X}} f_{Y|X}(y|x) P_X(x).$$

⑤ Maximum Likelihood estimators (ML).

$$\hat{x}_{\text{ML}}(y) = \arg \max_{x \in \mathcal{X}} f_{Y|X}(y|x).$$

Example from Quiz 4

$$X = \begin{cases} -2 & \text{w.p. } 1/4 \\ 2 & \text{w.p. } 3/4 \end{cases}$$

$$Y = X + Z \quad Z \sim N(0, 4).$$

$$f_{Y|X}(y|x) = f_Z(y-x)$$

$$Y|X=-2 \sim N(-2, 4).$$

$$Y|X=2 \sim N(2, 4).$$

a) MAP estimator

$$\hat{x}_{\text{MAP}}(y) = \begin{cases} -2 \\ 2 \end{cases}$$

$$\frac{e^{-\frac{1}{2}\left(\frac{y+2}{2}\right)^2}}{\sqrt{2\pi} \times 2} \times \frac{1}{4} \geq$$

$$f_{Y|X}(y|-2)P_X(-2) \geq f_{Y|X}(y|2)P_X(2)$$

otherwise.

$$\frac{e^{-\frac{1}{2}\left(\frac{y-2}{2}\right)^2}}{\sqrt{2\pi} \times 2} \times \frac{3}{4}$$


$$\frac{1}{3} \geq e^{\frac{1}{2} \left[\frac{(y+2)^2 - (y-2)^2}{4} \right]}$$

$$= e^{\cancel{\frac{1}{2}} \times \frac{\cancel{4} \times y \times \cancel{2}}{\cancel{4}}}$$

$$\Leftrightarrow y \leq \ln\left(\frac{1}{3}\right).$$

$$\hat{x}_{\text{MAP}}(y) = \begin{cases} -2 & y \leq \ln\left(\frac{1}{3}\right) \\ 2 & y > \ln\left(\frac{1}{3}\right). \end{cases}$$

⑥

ML estimate

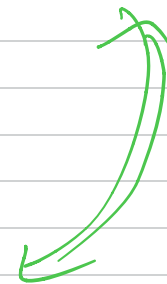
$$\hat{x}_{\text{ML}}(y) = \begin{cases} -2 \\ 2 \end{cases}$$

$$f_{Y|X}(y|-2) \geq f_{Y|X}(y|2)$$

Otherwise

$$e^{-\frac{1}{2} \left(\frac{y+2}{2} \right)^2}$$

$$\geq e^{-\frac{1}{2} \left(\frac{y-2}{2} \right)^2}$$



$$1 \geq e^{\frac{1}{2} \left(\frac{y+2}{2} \right)^2 - \left(\frac{y-2}{2} \right)^2} = e^y$$

$$\Leftrightarrow y \leq 0.$$

$$\hat{x}_{ML}(y) = \begin{cases} -2 & y \leq 0 \\ 2 & y > 0 \end{cases}.$$

Prob of estimation error for MAP and ML estimators.

② MAP estimation error.

$$\begin{aligned} P(X \neq \hat{x}_{MAP}) &= P(X \neq \hat{x}_{MAP}(Y)) \\ &= P(X \neq \hat{x}_{MAP}(Y) | X = -2) P(X = -2) \\ &\quad + P(X = 2) P(X \neq \hat{x}_{MAP}(Y) | X = 2) \\ &= P(\hat{x}_{MAP}(Y) \neq -2 | X = -2) \frac{1}{4}. \end{aligned}$$

$$+ P(\hat{x}_{\text{MAP}}(Y) \neq 2 | X=2) \frac{3}{4}.$$

$$= P(Y > \ln(\frac{1}{3}) | X=-2) \frac{1}{4}.$$

$$+ P(Y \leq \ln(\frac{1}{3}) | X=2) \frac{3}{4}.$$

$$P(Y > \ln(\frac{1}{3}) | X=-2) = P(X+Z > \ln(\frac{1}{3}) | X=-2).$$

$$= P(Z > 2 + \ln(\frac{1}{3}) | X=-2).$$

Z and X are independent $\leftarrow$

$$= P(Z > 2 + \ln(\frac{1}{3})).$$

$$Z \sim N(0, \sigma^2=4).$$

$$= P\left(\frac{Z}{2} > 1 + \frac{1}{2} \ln(\frac{1}{3})\right).$$

$$= 1 - \Phi\left(1 + \frac{1}{2} \ln(\frac{1}{3})\right) = 1 - \Phi(0.45)$$

$$P(Y \leq \ln(\frac{1}{3}) | X=2) = P(2+Z \leq \ln(\frac{1}{3}) | X=2).$$

$$= P\left(Z \leq \ln\left(\frac{1}{3}\right) - 2\right).$$

$$= \Phi\left(\frac{1}{2} \ln\left(\frac{1}{3}\right) - 1\right) = \Phi(-1.54).$$

$$P(X \neq \hat{x}_{MAP}) = \left(1 - \Phi(0.45)\right) \frac{1}{4} + \frac{3}{4} \underbrace{\left(\Phi(-1.54)\right)}_{\substack{\text{"} \\ (1 - \Phi(1.54))}}$$

$$= 1 - \left[\Phi(0.45) \times \frac{1}{4} + \frac{3}{4} \times \Phi(1.54)\right].$$

$$= 1 - \left[\frac{0.67364}{4} + \frac{3}{4} \times 0.93822\right].$$

$$= 1 - 0.872 = \boxed{0.1279}.$$

⑥ ML estimation error.

$$P(X \neq \hat{x}_{ML}) = P(\hat{x}_{ML}(Y) \neq -2 | X = -2) \frac{1}{4} + P(\hat{x}_{ML}(Y) \neq 2 | X = 2) \frac{3}{4}.$$

$$= P(Y > 0 | X = -2) \frac{1}{4} + P(Y \leq 0 | \underline{X=2}) \frac{3}{4}.$$

$X+Z \leq 0$

$$= P(Z > 2) \frac{1}{4} + P(Z \leq -2) \times \frac{3}{4}$$

$$= (1 - \Phi(1)) \frac{1}{4} + (\Phi(-1)) \frac{3}{4}.$$

$\Phi(-1) = 1 - \Phi(1)$

$$= 1 - \Phi(1).$$

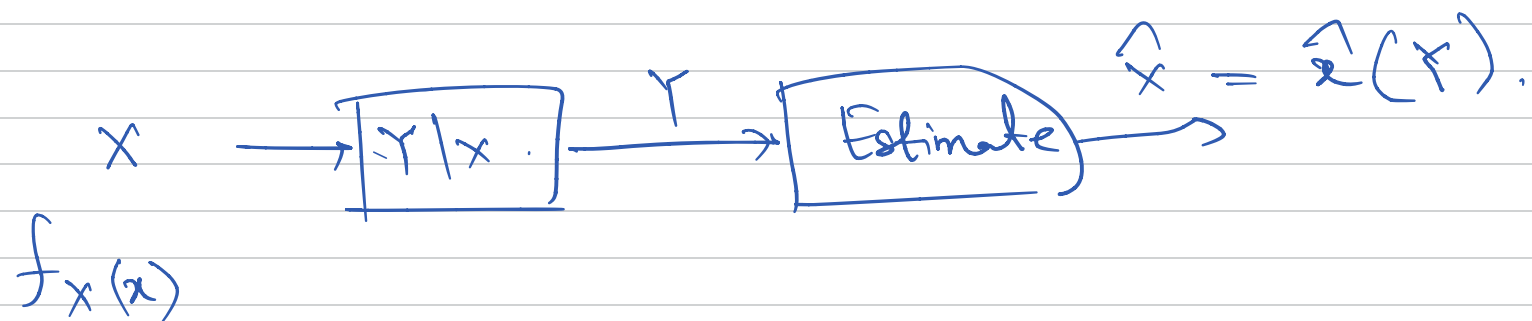
$$= 1 - 0.84134$$

$$\approx \boxed{0.16}$$

MAP has smaller estimation error than ML.

②

X is continuous



$$\text{Mean Square Error} = E[(X - \hat{X})^2].$$

③ Suppose $\hat{x}(Y)$ is a "constant" estimator.

$$\hat{x}(y) = c \quad \forall y \in \mathbb{R}.$$

$$\begin{aligned} \min_c E[(X - c)^2] &= \min_c E[X^2 - 2Xc + c^2] \\ &= \min_c \underbrace{E[X^2] - 2E[X]c + c^2}_{f(c)} \end{aligned}$$

$$\frac{\partial f(c)}{\partial c} = 2c - 2E[X] = 0 \Rightarrow c^* = E[X]$$

$$\frac{\partial^2 f(c)}{\partial c} = 2 > 0 \Rightarrow c^* \text{ is the minimum.}$$

The best possible (in mean square error (MSE)) constant estimator is $\hat{c}(Y) = E[X]$.

Example: $X \sim \text{Uniform}[4, 10]$ What is the best constant MNSE estimator.

$$\hat{c}_{\text{BCE}}(Y) = E[X] = \frac{10+4}{2} = 7.$$

Best Constant estimator

$$\begin{aligned} \text{MSE error of BCE estimator} &= E[(X - \hat{c}_{\text{BCE}}(Y))^2] \\ &= E[(X - E[X])^2] \\ &= \text{Var}(X). \end{aligned}$$

$$\text{Var}(X) = \int_4^{10} (x-7)^2 f_X(x) dx.$$

$$= \int_4^{10} \frac{1}{6} (x-7)^2 dx.$$

$$y = x-7.$$

$$= \int_{-3}^3 \frac{1}{6} y^2 dy$$

$$= \frac{y^3}{3 \times 6} \Big|_{-3}^3.$$

$$= \frac{\cancel{2} \times 3^3 - 3^3}{\cancel{3} \times \cancel{6}}$$

MSE error seen by BCE is 3.

⑥ Estimating x from Y (Minimum MSE estimator)
MMSE estimator.

Goal is to
find
 $\hat{x}$ that

$$\min_{\hat{x}(\cdot)} E[(x - \hat{x}(Y))^2].$$

We'll show
that

$$\hat{x}_{\text{MMSE}}(y) = E[x | Y=y].$$

$$E[(x - \hat{x}(Y))^2] \geq E[(x - \hat{x}_{\text{MMSE}}(Y))^2].$$

$\Rightarrow \hat{x}_{\text{MMSE}}(y)$ indeed minimizes the MSE.

Proof: $E[(x - \hat{x}(Y))^2] = E[E[(x - \hat{x}(Y))^2 | Y]]$

$E[g(x, Y)] = E[E[g(x, Y) | Y]]$ follows from

$$= \int f_Y(y) E[(x - \hat{x}(y))^2 | Y=y] dy$$

BCE is
the mean.

$$\int f_Y(y) \cdot E[(x - E[x|Y=y])^2 | Y=y] dy$$

$$E[(x - \hat{x}(y))^2 | Y=y].$$

$$= E[x^2 - 2x\hat{x}(y) + \hat{x}(y)^2 | Y=y]$$

$$\hat{x}^*(y) = \arg \min_c \underbrace{E[x^2 - 2xc + c^2 | Y=y]}_{f(c)}.$$

$$f(c) = E[x^2 | Y=y] - 2E[x | Y=y]c$$

$$\frac{\partial f(c)}{\partial c} = 2c - 2E[x | Y=y] \stackrel{!}{=} 0$$

$$\Rightarrow c^* = E[X|Y=y]$$

$$\Rightarrow \hat{x}^*(y) = E[X|Y=y].$$

$$\text{minimizes } E[(X - \hat{x}(y))^2 | Y=y]$$

$$\Rightarrow E[(X - \hat{x}(y))^2 | Y=y] \geq E[(X - E[X|Y=y])^2 | Y=y]$$

$$\Rightarrow \int f(y) E[(X - \hat{x}(y))^2 | Y=y]$$

$$\geq \int f(y) E[(X - \hat{x}_{\text{MMSE}}(y))^2 | Y=y]$$

$$\Rightarrow E[(X - \hat{X})^2] \geq E[(X - \hat{X}_{\text{MMSE}})^2].$$

$$\hat{x}_{\text{MMSE}}(y) = E[x|Y=y].$$

Example:

$$X \sim \text{Uniform}[4, 10].$$

$$W \sim \text{Uniform}[-1, 1]. \quad (\text{noise}).$$

$$Y = X + W$$

$$\hat{x}_{\text{MMSE}}(y) = E[x|Y=y].$$

Find $f_{X|Y}(x|y)$. and then find $E[x|Y=y]$.

$$\hat{x}_{\text{BCE}}(y) = E[x] = 7.$$